

Acoustic Triangulation Device

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Abstract — The Acoustic Triangulation Device (ATD) is an electronic system designed to detect the location of a sonic event, specifically gunshots or explosions, and relay that location in real time to the user. Its main subsystems include but are not limited to a microphone array, GPS locator, and real time triangulation software. These systems will work together to determine the origin of any sonic event within a predetermined frequency range. This relative location will be translated into GPS coordinates and relayed a desired remote device.

Index Terms — acoustic position measurement, acoustic propagation, wavelet transform, stochastic approximation.

I. INTRODUCTION

The Acoustic Triangulation Device (ATD) will consist of two microphone arrays connected to a computer. Each microphone array consists of three microphones in an equilateral triangle configuration. The microphones will be connected to a central timing unit that will measure the exact time that each microphone detects a valid event. The accuracy of the unit highly depends on the speed of our clock and the geometry of the array. Both the geometry of the triangle and the arrival time of the events will be used by the triangulation software to calculate a unit vector in the direction of the event's origin. A second array will simultaneously time the event and the two calculated unit vectors will intersect at the acoustic source.

A Global Positioning System (GPS) unit will be present in the central unit. This will provide the triangulation software with the absolute position of each array and allow for central reference points in our calculations. These reference points will then be coupled with the relative vectors produced by the microphone array to provide an absolute position of the source. The source location can then be transmitted to the authorities, emergency services, or other remote user enabling them to take appropriate action in a timely manner.

The real time triangulation software will act as a central hub for all sensor array and GPS information. The event

times for each speaker array will be processed through a triangulation algorithm and a unit vector in the direction of the source will be calculated. The intersection of the unit vectors from each array will be calculated to provide the source location relative to the sensor location. The software will then use this relative location as well as the absolute location of each array to calculate the absolute coordinates of the source. The source coordinates will be displayed in the user interface along with an alert letting the user know an event has occurred and a map of the event's location. User options will include relaying step by step directions to the source, saving the source sound waves in a database for later playback, and transmitting the source coordinates to a remote user. The software will be designed to install and run in a Microsoft Windows environment.

The ATD will also be able to analyze and distinguish between types of events. The wavelet transforms will be used to compare an event to a database of event wavelets and provide information about the source to the user such as gunshot type.

II. TRIANGULATION

Initially we tried to implement multilateration. Eventually we found out that multilateration required more expensive and sensitive equipment. We also discovered that it required a stochastic gradient regression method to properly estimate the source location. After having much difficulty with multilateration we decided to derive our own equations using triangulation instead. For two-dimensional triangulation we need two arrays of three microphones each, oriented in an equilateral triangle, to determine the exact location of a sound source. Each array will give us an angle which tells us the direction of the source. Using these two directions and the known distance between the two arrays we can determine the sound source's location.

If the microphones are close enough together and the sound source is sufficiently far away we can assume the sound wave approaches as a straight line perpendicular to the line originating from the source. We can then find the distance Δx , using the distance/rate/time formula where the distance is Δx , the rate is the speed of sound, C , and the time is the difference between the time of first detection and the time of second detection, $t_B - t_A$.

$$\Delta x_1 = C \times (t_B - t_A)$$

The speed of sound, C , changes in relation to the temperature of the air. Other factors can affect the speed of sound, such as the barometric pressure and humidity,

however these factors are insignificant in comparison to the temperature. The temperature, T , is measured in $^{\circ}\text{C}$.

$$C = 331.3 + .606 \times T$$

Knowing the distance, Δx , and the side of the array, S , we can find the angle θ_1 using trigonometry. Then we can find the angle α_1 based on its relationship with θ_1 .

$$\theta_1 = \cos^{-1}\left(\frac{\Delta x_1}{S}\right), \quad \alpha_1 = \theta_1 - 30^{\circ}$$

These equations will work regardless of the orientation of the array, therefore the times t_A and t_B will always be the time that the first and second microphones detect a sonic event, respectively. Based on these equations there are two locations that the source could come from, one on each side of the line which connects the first and second microphones. The third microphone tells us that the source came from the opposite side that it is located.

There are two equations which can tell us the value of Δy_1 . The first is based on the speed of sound and the times of the second and third detections of the sonic event. The second equation is based on the triangle made up of Δy_1 and the base of the array, and the angle determined by the previous equations.

$$\begin{aligned} \Delta y_1 &= C \times (t_C - t_B) \\ \Delta y_1 &= S \times \cos(180^{\circ} - 60^{\circ} - \theta_1) \end{aligned}$$

If these two equations are not equal, then there is some error involved in the calculations. This error could be caused by the source being at a location other than ground level. Alternatively the error could be caused by inaccuracies in the time readings.

All of the above equations that were used to find the values of the first array, and therefore the angle α_1 , can also be used to find the values of the second array, and therefore the angle α_2 . Using the angles α_1 and α_2 found by the previous equations, we can determine the angles β_1 , β_2 , and β_3 of the larger triangle formed by the lines connecting the two arrays and the sound source.

The relationship between the β angles and the α angles will have to be determined by knowing the orientation of each array with respect to the line that connects the two arrays. This information will be determined using a compass on each array. Based on the case in which both arrays are in the same orientation and their bases are parallel to the line that connects them to each other, the correct equations to use would be the following equations.

$$\begin{aligned} \beta_1 &= 90^{\circ} + \alpha_1 \\ \beta_2 &= 90^{\circ} - \alpha_2 \\ \beta_3 &= 180^{\circ} - (\beta_1 + \beta_2) \end{aligned}$$

Then using the Law of Sines we can determine the distance from the first array to the sound source.

$$D = \frac{\sin \beta_2 \times L}{\sin \beta_3}$$

In order to find the exact location, we then need to know the exact location of each array. This information is given to us by the GPS. Since the size of each array is small in comparison to the distance from the sound source to the arrays, the GPS can be located at any point inside the array and still give a good enough approximation of the array's location. This means that we can say that each microphone is at approximately the same location as the GPS unit.

The coordinates for the sound source are found by adding the vertical portion of the distance to the vertical coordinate of the GPS and the horizontal portion of the distance to the horizontal coordinate of the GPS for the first array. The vertical and horizontal directions will have to be normalized to North/South and East/West directions to find the proper coordinates. This will be accomplished by using the compass values for each array and will adjust the α and β angles accordingly. The equations for finding the vertical and horizontal portions of the distance to the sound source based on the case in which the positive vertical direction is North and the positive horizontal direction is East, are the following.

$$\begin{aligned} D_{\text{vert}} &= D \times \sin(180^{\circ} - \beta_1) \\ D_{\text{horiz}} &= D \times \cos(180^{\circ} - \beta_1) \end{aligned}$$

Combining these equations we can get a single equation for the angle that each array produces with only the variables t_A , t_B , the temperature, T , and α .

$$\begin{aligned} \alpha_1 &= \cos^{-1} \frac{(331.3 + .606 \times T) \times (t_{B1} - t_{A1})}{S} - 30^{\circ} \\ \alpha_2 &= \cos^{-1} \frac{(331.3 + .606 \times T) \times (t_{B2} - t_{A2})}{S} - 30^{\circ} \end{aligned}$$

We can also combine the previous equations to get a single equation for the distance between the sound source and the first array, D , with only the variables α_1 , α_2 , the distance between the two arrays, L , and D .

$$D = \frac{\sin(90^{\circ} - \alpha_2) \times L}{\sin(180^{\circ} - ((90^{\circ} + \alpha_1) + (90^{\circ} - \alpha_2)))}$$

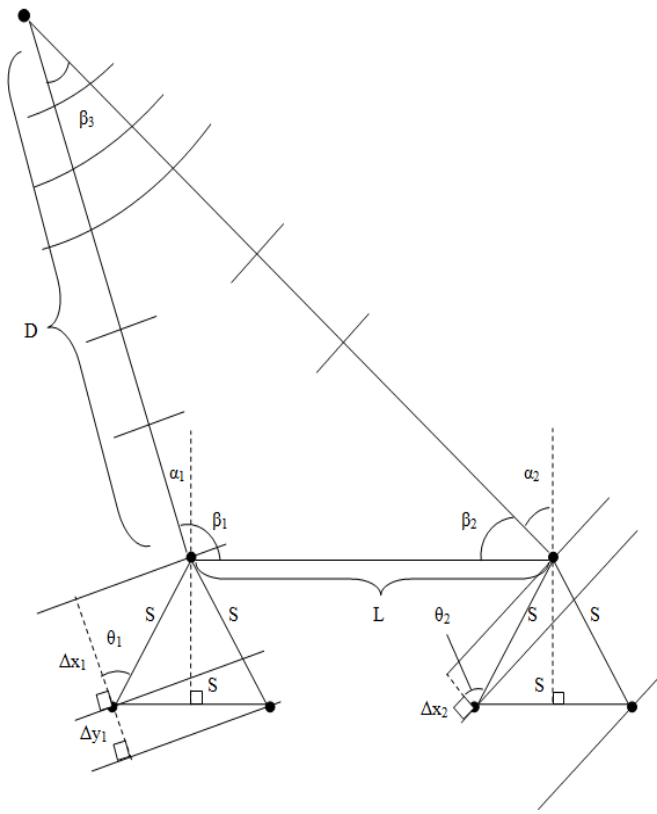


Figure 1 – Microphone array configuration

III. WAVELETS

Wavelets are localized waves whose energy is concentrated in time and space and are perfect for the analysis of transient signals. A wavelet transform is the representation of functions using wavelets. Wavelets are scaled into copies of daughter wavelets over a finite length non periodic waveform referred to as the mother wavelet. Wavelets are better used than Fourier analysis for our project, because they are used for non periodic waveforms, and they are also ideal at representing sharp peaked functions, such as the characteristic of a gunshot.

The type of wavelet transformation that we are interested in using for the ATD is the Discrete Wavelet Transform. The DWT is easy to implement and has a fast computation time with minimum resources required. In order to get the DWT, the use of high pass and low pass filtering is used on the signal.

Down sampling is used in DWT because for every time you filter, you are incrementing by a large amount of data, so it is necessary for down sampling to occur. It should also be noticed that when half of the frequencies of the

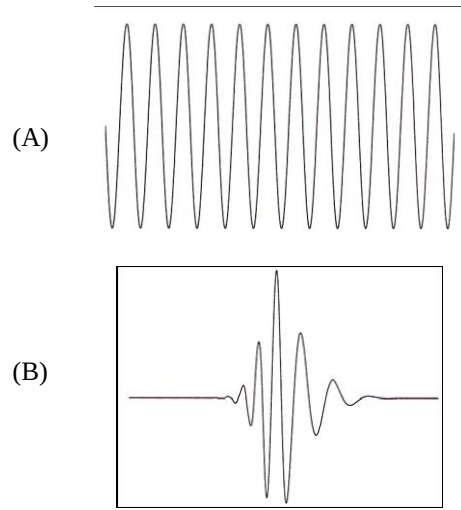


Figure 2 – Demonstration of a (A) wave and a (B) wavelet

signal are removed, half the samples can be discarded according to Nyquist's rule. After this entire process is completed, there will be numerous signals that represent the same signal, but each signal will correspond to a specific frequency range. This process can be repeated multiple times, and the number of times it is repeated corresponds to what the application calls for. For our design, we determined that level 4 wavelet transform was the most appropriate.

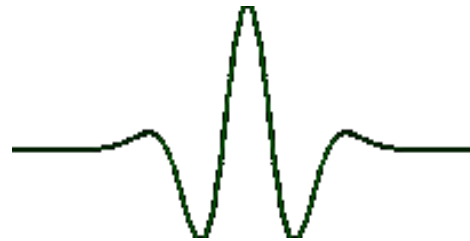


Figure 3 – Gaussian Mother Wavelet

There are many different types of wavelet shapes to choose from. Using Matlab we have come across the Daubechies wavelet. This wavelet is most similar in shape to that of a gunshot. The following figure is an example of the Daubechies wavelet in Matlab.

The figure below shows a Daubechies wavelet in Matlab with a decomposition level of four. As you can see from the figure, it shows the decomposition of the low and high pass filter. This wavelet will be used in our project to divide our time signals into different scaled components. After each component is scaled, we will be studying each component. This data compression

technique will then allow us to compare our event gunshots, with our database of stored characteristics.

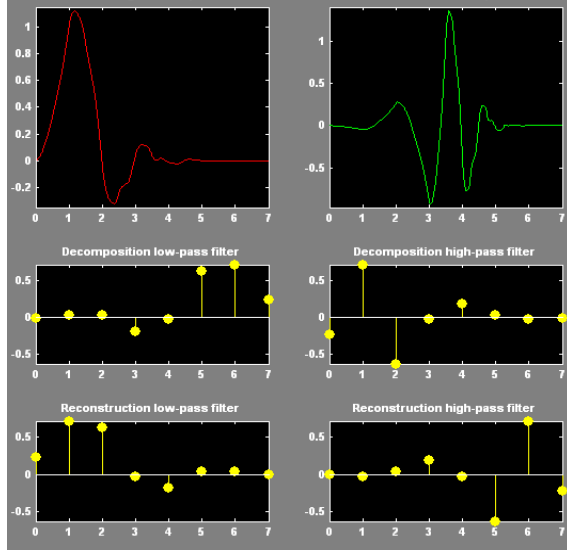


Figure 4 a) – Daubechies Wavelet level 4

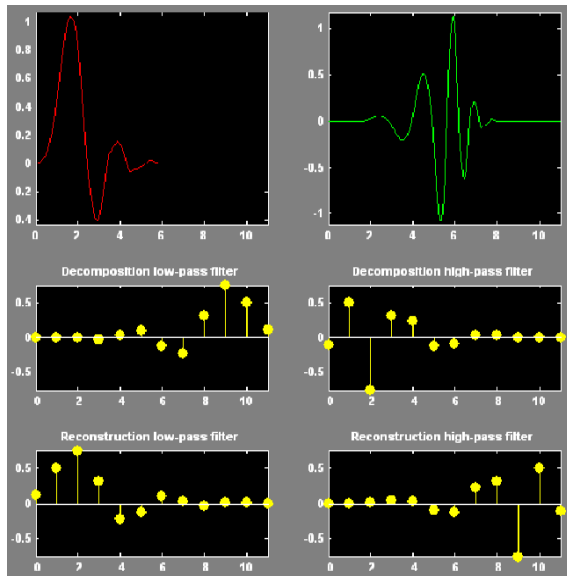


Figure 4 b) – Daubechies Wavelet level 6

The above figure shows a Daubechies wavelet in Matlab with a decomposition level of six. As you compare the difference of decomposition levels, you can notice the wavelet functions difference. The higher the decomposition level, the more compressed the signal becomes. The Daubechies family can either be orthogonal or biorthogonal.

Equations for the discrete wavelet transform:

$$W(j, k) = \sum_j \sum_k x(k) 2^{-\frac{j}{2}} \varphi(2^{-j} n - k)$$

In the above equation $\varphi(t)$ is the mother wavelet. This mother wavelet is a representation of a function in time with a finite energy. This mother wavelet also has a fast decay time. The discrete wavelet transform can be computed using an extremely quick pyramidal algorithm. This pyramidal algorithm allows the signal to be analyzed in different octave frequency bands and it allows different resolutions by decomposing the signal in coarse approximations and detail information. This decomposition as mentioned above is done by using the high pass and low pass filtering of the time domain signal.

$$Y_{high}[k] = \sum_n x[n]g[2k - n]$$

$$Y_{low}[k] = \sum_n x[n]h[2k - n]$$

Where $Y[k]$ are the outputs of the high and low pass filters after the down sampling of a factor of two. The wavelet coefficients represent the energy of the signal in the time and frequency. These coefficients can be represented by using different techniques such as by taking the ratios of the mean values between adjacent sub bands in order to provide information on the frequency distribution. Other techniques such as by taking the mean value of the coefficients in each sub band can also provide information on the frequency distribution. In order to get the change in frequency distribution, we must take the standard deviation in each sub band. For our design purposes, we will be comparing the coefficients of the wavelet transforms.

To make certain that all of our wavelet transforms are comparable to each other, all of our scales of our wavelet functions must be normalized to have unit energy. The equation we will be implementing to do the normalization is as follows:

$$\hat{\varphi}(s\omega_k) = \left(\frac{2\pi s}{\delta t}\right)^{\frac{1}{2}} \hat{\varphi}_o(s\omega_k)$$

Then, after normalization each scale s has

$$N = \sum_{i=0}^K |\hat{\varphi}(s\omega_k)|^2$$

with N being the number of points. By using the convolution formula, normalization of our functions to have unit energy can be done by:

$$\varphi \left[\frac{(n' - n)\delta t}{s} \right] = \left(\frac{\delta t}{s} \right)^{1/2} \varphi_o \left[\frac{(n' - n)\delta t}{s} \right]$$

Since in our design we have chosen to use the Daubechies wavelet function, which is an orthogonal wavelet function reconstruction of the original time signal can be determined using deconvolution.

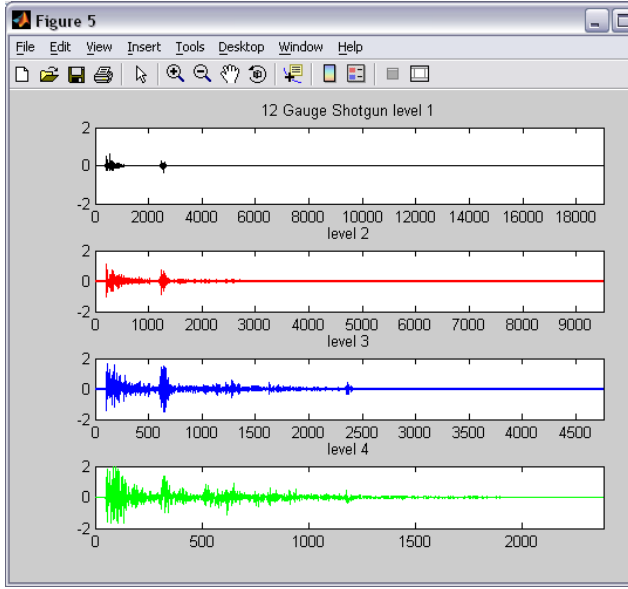


Figure 5 – Wavelet transform of a shotgun

The very first step in process of different type of gunshot recognition by using the discrete wavelet transform will be to take our recordings of the various types of guns that we have shot, eliminate all noises that do not pertain to the exact gunshot, and then to normalize the gunshot signatures. The reason that we will be normalizing these recordings is so that the length of each type of gunshot does not affect our results. After we do the normalizing process, we will then proceed to do the discrete wavelet transform to each of the gunshot recordings.

The next step will then be to take each of the gunshot recordings, and get the absolute value of the coefficients. We will then begin the storing process on these discrete wavelet transforms and coefficients into a database in the computer. After the storing process is complete, we will then be able to take the gunshot events, and do the process in order to distinguish them in real time. After we have the normalized signal, the discrete wavelet transform will be done on the signal.

We will be first taking the input signal of the gunshot events from the microphone that has been sent to Matlab

by way of the ATD software and normalize them. The next process will be to store these events into a database in the computer. After the storing process is complete, we will then proceed and compare the event gunshots with our database of stored recordings in the computer. After the gunshot comparison of the coefficients and signals, we will then use our tolerance algorithm to output the best match of the type of gun that was used from the gunshot event.

V. DESIGN SUMMARY

In an outdoors environment the ATD will be standalone, most likely solar powered with its own LCD display and keypad. These components have been omitted from the block diagram as they are not design critical. Also not shown is the ATD to computer connection that will be present if the unit is not standalone. This connection however is design critical and will use the same USB connection the power comes from. The Arduino's USB connection will be a two way communication network between the user and ATD as well as a convenient power supply.

Each of the six microphones and their corresponding break out boards will be connected to the Arduinos 5V voltage output to supply power. The ATD will by default be in listening mode and the microphones will be sampled at 2 MHz. The analog signal representative of the sound source is taken in through each of six microphones. This signal is then amplified to a useable voltage by each microphone's corresponding break out board. The amplified signal is processed through six of the Arduino's sixteen analog inputs and converted through the board's analog to digital converter (ADC). The samples from the Arduino will not need to be stored on the board because the samples will be evaluated individually to see if they go above the threshold for the sound of a gunshot. Once a microphone detects that it is over the threshold the time will be stored. Then each subsequent microphone will also store the time when it samples a voltage that is above the threshold. These times are the times of arrival that will be used by the triangulation subroutine. Due to the nature of finding the first peak in the sound wave which is above the threshold of a gunshot, the Nyquist Shannon sampling theorem is a minimum. It will be helpful to sample as fast as possible to ensure that a peak is found, without having to reconstruct the wave. The minimum sample rate, based on the Nyquist Shannon theorem, for a 4000 Hz sound wave (which is the highest frequency in an average gunshot) is 8000 Hz. We will be sampling at 2 MHz so we are well beyond this minimum. Once the event is detected, the board will send a signal to the computer to start recording. In this way we will be able to store more information because the computer has significantly more

memory than the Arduino. This recording will be used for distinguishing gunshots using wavelet transforms.

The thermometer is also connected to the voltage source as well as the RX1 serial input on the Arduino Mega. The current temperature will be stored in the board's memory and updated every ten seconds. This way if a cloud moves overhead or the temperature drops suddenly the unit will still maintain accuracy. Temperature affects the speed of sound very profoundly and is not to be overlooked. The current temperature will be used to calculate the speed of sound more accurately and this information will also be sent to the triangulation subroutine.

The digital compass (also powered on the 3V3 Arduino output) has a clock which is connected to the clock regulating signal SCL on the Arduino board. This signal allows the output input from the compass to sync with the input on the Arduino. The high/low signal SDA is then transmitted to digital pin 22 for storage and the value is decoded then used in reference frame setup in the triangulation subroutine. Without the digital compass it would be impossible for the ATD to give the coordinates of an event and as such it is imperative that the compass transmits an accurate bearing at setup time.

The GPS is again powered on the 3V3 line. As a side note the sum of all amperages specifications of the components does not exceed the Arduinos 3V3 output specifications. The GPS transmit line is connected to the RX0 serial input on the Arduino Mega and transmits a NMEA 0183 standard string. This string is then decoded by the GPS decode subroutine and the GPS coordinates of the ATD are extracted. These GPS coordinates are then sent to the Triangulation subroutine and used to set the reference frame from which to calculate the target event's GPS location as follows.

The variable for temperature is initialized and the speed of sound is calculated by using this formula

$$C = 331.3 + .606 \times T$$

The variable for the event arrival time from each of the six analog inputs is passed into each of these equations and the coordinates relative to our predefined x and y axis are calculated. The x and y axis are defined by the north, south, east, and west provided by the digital compass and the origin is defined to be the ATD coordinates provided by the GPS. Note that each microphone will have its own coordinates as well calculated by the ATD's dimensions in conjunction with the physical GPS location onboard the unit. Once the reference frame is set the calculation variables are initialized from microphones, temperature sensor, compass, and GPS, and the source location is triangulated then passed to the user.

Once the exact event location has been calculated the signal analysis subroutine takes over and analyzes the

wavelet breakdown of the event. A predefined set of wavelets will be saved into the computer's memory bank and the current waveform will be compared with all previously saved waveforms. The wave that most closely matches the current wave will provide the signal analysis routine with the initial type of explosive, caliber of bullet or type of gun of the event. The user will have the option of classifying and storing a new wavelet for future acoustic even analysis.

An additional option for scalability would be the ability of the ATD to store and classify its own new wavelets based on what it learns from previous information. If a close enough match is not found the ATD will store the wavelet with an arbitrary name and wait for an external input to give it a classification. In the time between storage and classification the ATD would still be able to match new event waves to this arbitrary waveform. The map locations and types of events will be stored in an ongoing record until the ATD is reset. This will become useful in investigations, steak outs, and case studies of crime in certain areas where event patterns may emerge that would be useful to certain authorities.



Figure 6 – Map of located gunshot

The Arduino will then transmit this information to an external monitor or to a computer where the information is further analyzed. If the information is sent to the computer a map of the GPS location will be pulled up using Google maps, a screen shot will be taken and a picture of the round, explosion or gun type will be inlayed into the correct position on the said map. The wave sound will also be provided to the user in .wav format so that they may intuitively be able to verify the information the ATD is providing them with and reclassify the acoustic event as necessary. Note reclassifying wave events will affect the functionality of the ATD and must be presented only as an advanced option for the user.

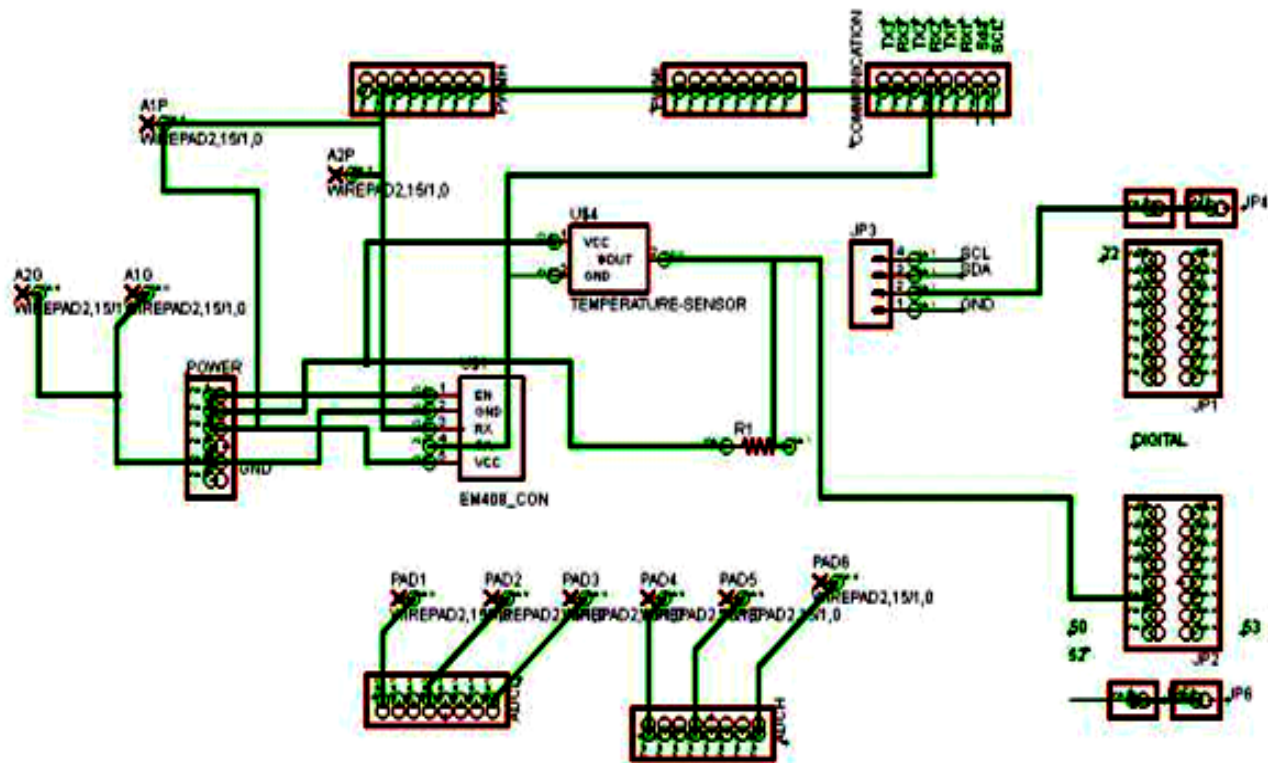


Figure 7 – ATD block diagram

VII. CONCLUSION

Once the process is complete the user will be given the option to save the event which will save the map image, wave file, longitude and latitude into a single folder, or send the event via email or other medium to the correct authorities. The user may also view all events stored of the current area on one map to reveal event patterns. There will be search options which would allow the user to search by area, event type, or event frequency allowing the user pull up information such as which locations have the most gunfire. When the processing is complete the ATD returns to listening mode once again.

Additional options for scalability will include and LCD display on the unit which will display the latitude and longitude of the event as well as the type. Also the unit may have a keypad. This coupled with a power source(solar, battery, other) will allow the unit to be standalone and only raise the price by a little under \$100. The current version of the ATD is a prototype which most likely will not be standalone and as such the above additions are beyond the scope of this paper.

The ATD uses many concepts that we have learned in our Electrical and Computer Engineering classes at the University of Central Florida.

We believe that our ATD prototype has many useful applications that it can be implemented with. In the beginning, we were focused on working on a project that we would all be interested in. The ATD was exactly what we were looking for in a design project.

The version of the ATD presented in this paper is best suited for high profile events such as political speeches. The unit is portable and easy to calibrate and set up quickly. It is durable enough to withstand the packing and unpacking from storage but is not suited for extended outdoor use. It can locate targets accurately at a limited distance of 400 meters and as such is suited for the speech setting. It must be powered by and used from a computer or a laptop which will be readily available at such events. This is a first prototype and all tests were completed in an open field.

The cost of our ATD design was extremely impressive. In the beginning, we believed that this project would cost around a total of one thousand dollars. From our research we have learned that the total cost of the ATD will be approximately three hundred dollars. This is extremely nice because all of the current gunshot detectors are very expensive to purchase.

The ATD has been a fun and exciting experience as our first design project. Not only was the research and design of the ATD fun and exciting, and while working on the ATD we have learned quite a lot. We have learned about things such as triangulation, multilateration, signal analysis, filtering and how each of the different components needed for the ATD to work. We believe that the designing processes overall went very well, and that we worked incredibly hard to get this design completed. We believe that our vast research helped to ensure that that the ATD became a successful project.

ACKNOWLEDGEMENT

We would like to thank Robi Polikar from Rowan University author of The Wavelet Tutorial. Special thanks to him for allowing us to use pictures from his website at <http://users.rowan.edu/~polikar/WAVELETS/WTtutorial.html>. Also a special thanks to Louis Scheifereeder for allowing us to use several of his firearms for testing purposes.

BIOGRAPHY



Jeremy Hopfinger, a senior student of the electrical engineering department at University of Central Florida, plans on starting his working career in the electrical engineering profession and eventually continue studies towards MSEE. Current research interest is in communications.



Johnathan Sanders is currently a senior at the University of Central Florida. He plans to graduate with his Bachelor's of Science in Computer Engineering in Spring of 2010. He is currently working, as a computer technician at Best Buy's Geeksquad. He plans to attend the University of Central Florida for his graduate studies in the field of Intelligent System Design.



Benjamin Noble is currently a senior at the University of Central Florida, and will graduate with a bachelor's degree in Computer Engineering in May 2010. He is currently employed as a Computer Engineer at CAE USA inc. located in Orlando FL. He plans to continue his education and eventually earn a Master's degree in Computer Engineering with a concentration on Artificial Intelligence.

REFERENCES

- [1] Rob Maher. (2007, April) Acoustical Characterization of Gunshots. [Online]. http://www.microflow.com/data/SSA_maher_ieeesafe_04_07_109-113.pdf
- [2] J. Hartikka.(1992) .308 Measured. [Online]. <http://guns.connect.fi/rs/308measured.html>
- [3] How Do Microphones Work?. [Online]. <http://www.mediacollege.com/audio/microphones/how-microphones-work.html>
- [4] National Semiconductor. [Online]. <http://www.national.com>
- [5] Our Approaches to the Project. [Online]. <http://www.ownet.rice.edu/~elec431/projects97/Dynamic/approaches.html>
- [6] The Discrete Wavlet Transform. [Online]. http://www.dtic.upf.edu/~xserra/cursos/TDP/referencies/Par_k-DWT.pdf
- [7] Sparkfun Electronics. [Online]. <http://www.sparkfun.com>
- [8] Kel-Tec CNC Industries INC. [Online]. <http://www.kel-tec-cnc.com/>
- [9] Jim Lesurf. The Sampling Theorem Signal Reconstruction. [Online]. http://www.st-andrews.ac.uk/~www_pa/Scots_Guide/iandm/part7/page3.html
- [10] Cuthbert Nyack. (2005) Wide Band Pass Butterworth Filter. [Online]. <http://cnyack.homestead.com/files/afilt/afilt-Butterworthwbp.htm>